As an aside, the fact that pretty much all the hydrogen in the universe was produced in the first few minutes means that a pretty large fraction of what you and I are made of has been hanging around the universe in one form or another for almost as long as the universe has been here. You may have heard that "we are made of stardust" (or "star stuff" if you're Sagan), and this is absolutely true if we measure *by mass*. All the heavier elements in your body—oxygen, carbon, nitrogen, calcium, etc.—were produced later, either in the centers of stars or in stellar explosions. But hydrogen, while the lightest, is also the most abundant element in your body *by number*. So, yes, you hold within you the dust of ancient generations of stars. But you are also, to a very large fraction, built out of by-products of the actual Big Bang. Carl Sagan's larger statement still stands, and to an even greater degree: "We are a way for the cosmos to know itself."

THE SURFACE OF LAST SCATTERING

After Big Bang Nucleosynthesis, things in the infernoverse started to settle down, relatively speaking. At that point, the mix of particles was more or less stable, and would remain so until the time of the first stars, millions of years later. But for hundreds of thousands of years, the cosmos was still a hot, humming plasma composed mostly of hydrogen and helium nuclei and free electrons, with photons (particles of light) bouncing around between them.

Over time, the expansion of the universe gave all that radiation and matter room to spread out. I sometimes imagine experiencing this phase of the early universe like a journey from the center of the Sun outward, but instead of moving through space, you're moving through time. You start in the center of the Sun, where the heat and density are so high that atomic nuclei are fusing together to make new elements. The solar interior is opaque with light, with photons continually bouncing off electrons and protons so violently that it can take hundreds of thousands of years of constant scattering for a photon to reach the surface. Eventually, as you move outward, the plasma becomes less dense and light is able to travel farther between scatterings. At the surface, it can stream freely out into space.

In a similar way, a journey through time from the first few minutes of the universe to about 380,000 years later takes the entire cosmos from that hot